Python Dunder (Double Underscore) Methods Cheat Sheet

Dunder methods, also known as magic or special methods, are predefined methods in Python with names surrounded by double underscores (__like_this__). They enable operator overloading, custom behavior for built-in functions, and integration with Python's syntax.

Dunder Method	Description
__init__(self, ...)	Constructor; called when creating a new instance.
__del__(self)	Destructor; called when an object is about to be destroyed.
__str__(self)	Defines the human-readable string representation (str()).
__repr__(self)	Defines the official string representation (repr()).
__len__(self)	Returns length (len()).
__getitem__(self, key)	Called to retrieve an item using indexing.
__setitem__(self, key, value)	Called to set an item using indexing.
__delitem__(self, key)	Called to delete an item using indexing.
__iter__(self)	Returns an iterator object (for loops).
__next__(self)	Returns the next item from an iterator.
__contains__(self, item)	Implements membership check (in).
__call__(self, ...)	Makes an instance callable like a function.
__enter__(self)	Context manager entry method (with statement).
__exit__(self, exc_type, exc_val, exc_tb)	Context manager exit method.

Dunder Method	Description
__add__(self, other)	Implements addition (+).
__sub__(self, other)	Implements subtraction (-).
__mul__(self, other)	Implements multiplication (*).
__truediv__(self, other)	Implements true division (/).
__floordiv__(self, other)	Implements floor division (//).
__mod__(self, other)	Implements modulus (%).
__pow__(self, other)	Implements exponentiation (**).
__eq__(self, other)	Implements equality (==).
__ne__(self, other)	Implements inequality (!=).
__lt__(self, other)	Implements less than (<).
__le__(self, other)	Implements less than or equal (<=).
__gt__(self, other)	Implements greater than (>).
__ge__(self, other)	Implements greater than or equal (>=).
__bool__(self)	Defines truthiness (bool()).
__hash__(self)	Defines the hash value (hash()).
__copy__(self)	Defines behavior for copy.copy().
__deepcopy__(self, memo)	Defines behavior for copy.deepcopy().